Chapter 7, Problem 35P

Problem

Find the solution to the following differential equations:

(a) $\frac{dv}{dt} + 2v = 0, \qquad v(0) = -1\ \text{V}$

(b) $2\frac{di}{dt} - 3i = 0, \qquad i(0) = 2$

Step-by-step solution

Step 1 of 4

(a)

Consider the following data:

First order differential equation,

$$\frac{dv}{dt} + 2v = 0$$

Initial voltage value, $v(0) = -1\ \text{V}$

This is a first order differential equation. Since only the first derivative of v is involved.

Rearrange the terms.

$$\frac{dv}{dt} = -2v$$

$$\frac{dv}{v} = -2dt$$

Integrate both sides.

$$\int \frac{dv}{v} = -2\int dt$$

$$\ln v = -2t + \ln C$$

Here C is the integration constant.

$$\ln v - \ln C = -2t$$

$$\ln\left(\frac{v}{C}\right) = -2t$$

Take power of e.

$$v(t) = Ce^{-2t}$$

At $t = 0$, $v(0) = C = -1$

Step 2 of 4

The solution to the differential equation is,

$$\begin{aligned} v(t) &= -1e^{-2t} \\ &= -e^{-2t}\ \text{V} \end{aligned}$$

Therefore, the solution to the given differential equation, $v(t)$ is $\boxed{-e^{-2t}u(t)\ \text{V}}$.

Step 3 of 4

(b)

Consider the following data:

First order differential equation,

$$2\frac{di}{dt}-3i=0$$

Initial current value, $i(0)=2\text{ A}$

Rearrange terms.

$$\frac{di}{dt}=\frac{3}{2}i$$
$$\frac{di}{i}=\frac{3}{2}dt$$

Integrate both sides.

$$\int\frac{di}{i}=\frac{3}{2}\int dt$$
$$\ln i=\frac{3}{2}t+\ln C$$

Here C is the integration constant.

$$\ln(i-C)=\frac{3}{2}t$$
$$\ln\frac{i}{C}=1.5t$$

Take power of e.

$$i(t)=Ce^{1.5t}$$

At $t=0$, $i(0)=C=2$

Step 4 of 4

The solution to the differential equation is,

$$i(t)=2e^{1.5t}\text{ A}$$

Therefore, the solution to the given differential equation, $i(t)$ is $\boxed{2e^{1.5t}u(t)\text{ A}}$.